

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name	Rushda Khan
Program / Batch	B083-Networkwalks
Date	16 September 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client / Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3–5: In Progress

1. Liability Disclaimer

I have performed these activities only on systems and devices where I had secured written permission, or on devices/systems that I own myself. All materials referenced here are for education and research purposes only. I have not used anything from this project to break the law. The instructor, the authors and Networkwalks are not responsible for what I do with this knowledge outside this scope. Every action I take is my own responsibility. I understand that misuse can lead to criminal charges, heavy fines, loss of employment and a permanent record, and that in most countries unauthorized access is a crime even when nothing is damaged.

2. Introduction

This report covers the footprinting of the `networkwalks.com` domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network using Zenmap (W2-PM5). The first module focuses on the footprinting phase, while the second focuses on network scanning. Together, they helped me understand how information about a target can be gathered first and then how live devices on a network can be identified and mapped.

This is my Week 2 deliverable for my ongoing Cybersecurity & Ethical Hacking learning program.

All the footprinting commands were performed in Kali Linux, while the network scanning was done on my Windows PC using Zenmap. For each task, I have included the command used, the result I got, a screenshot as evidence, and a short explanation of why the finding is important from a cybersecurity perspective.

3. Tools Used

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance and scanning activities
WHOIS	Find domain registration details (owner, dates, name servers)
whatweb	Fingerprint web technologies (server, CMS, plugins, IP)
nslookup	Resolve the domain name to its IP address using DNS
curl -I	Read the HTTP response headers of the website
wafw00f	Detect whether a Web Application Firewall protects the site
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV)
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses
Windows CMD	Local IP and MAC address identification

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance on the `networkwalks.com` domain using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon. Each tool was used to collect a different type of information about the domain and website.

First, I used **WHOIS** to collect publicly available domain registration information and identify the domain's name servers. The results provided information about the domain registration, registrar and DNS infrastructure.

I then used **WhatWeb** to identify the technologies being used by the website. The results showed **WordPress 7.1** and **WordPress Download Manager 3.3.58**, along with other information exposed by the website.

Using **Nslookup**, I resolved the domain name to its IP address. The result showed that `networkwalks.com` resolves to **192.232.216.135**.

I used **Curl** with the `-I` option to view the HTTP response headers. This provided additional information about the web server and application, including the WordPress REST API endpoint `/wp-json/`.

Next, I used **Wafw00f** to check whether a Web Application Firewall (WAF) was being used by the website. The result identified **ModSecurity (SpiderLabs)**.

Finally, I used **DNSRecon** to enumerate the available DNS records. The results provided information about the **name servers, mail server, A record, SPF/TXT records, SRV records and DNS software information**.

Overall, these tools helped me collect different types of publicly available information about the domain and understand what information can be exposed during the footprinting and reconnaissance phase.

4.2 Network Scanning with Zenmap

For the second activity, I used **Zenmap** to perform network scanning on my own local network. The practical required me to find my local IP address and subnet, identify live hosts on the network, find their IP and MAC addresses, and generate a network topology.

I first used the Windows `ipconfig` command to find my local IP address and LAN subnet. My Windows PC had the IP address **10.0.0.10** with the subnet mask **255.255.255.0**, which gave me the subnet **10.0.0/24**.

I then entered this subnet into Zenmap and selected **Ping Scan** to discover the active hosts on the network. The scan identified **4 live hosts**: `10.0.0.1`, `10.0.0.2`, `10.0.0.3`, and `10.0.0.10`. The scan results also provided MAC address information for some of the detected hosts.

Finally, I used Zenmap to generate and save the **network topology in PDF format**, as required by the practical.

Command used:

```
nmap -sn 10.0.0.0/24
```

Actual result:

```
Starting Nmap 7.991 ( https://nmap.org ) at 2026-09-15 20:00 -0700
Nmap scan report for 10.0.0.1
Host is up (0.0030s latency).
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
Nmap scan report for 10.0.0.2
Host is up (0.0030s latency).
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
Nmap scan report for 10.0.0.3
Host is up (0.0029s latency).
MAC Address: 08:00:27:5A:87:BC (Oracle VirtualBox virtual NIC)
Nmap scan report for 10.0.0.10
```

```
Host is up.  
Nmap done: 256 IP addresses (4 hosts up) scanned in 3.24 seconds
```

The scan identified four live hosts on my local subnet:

- 10.0.0.1 — MAC 52:54:00:12:35:00 (QEMU virtual NIC)
- 10.0.0.2 — MAC 52:54:00:12:35:00 (QEMU virtual NIC)
- 10.0.0.3 — MAC 08:00:27:5A:87:BC (Oracle VirtualBox virtual NIC)
- 10.0.0.10 — my own PC (no MAC shown, since a host cannot ARP itself)

I noted that 10.0.0.1 and 10.0.0.2 returned an identical MAC address. This is not typical on a physical network, but it is a known behaviour in QEMU/KVM virtual lab environments, where the hypervisor can assign the same virtual MAC to more than one guest interface depending on how the virtual network is configured.

After completing the scan, I opened the Topology section in Zenmap, enabled the legend, and saved the network topology in PDF format as required by the practical task.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager	Attackers may use exposed technology/version information to identify software requiring further security review	Medium
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Provides information about the network location of the web service	Low
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/	May assist technology fingerprinting and further enumeration	Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	Low
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records	DNS information can help build a broader infrastructure profile	Medium
6	Multiple live hosts visible on local network	Zenmap identified four live hosts, including two sharing an identical MAC address	Unknown or unauthorized devices may potentially be present on a network; duplicate MACs may also indicate a virtualization/network configuration worth reviewing	Medium

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities. The practical exercises primarily involved information gathering and host discovery — no exploitation or vulnerability validation was performed as part of these two modules. Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that a system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I would recommend the following security improvements:

1. Review publicly exposed technology information

Organizations should regularly check what information about their web technologies, CMS, and plugins is publicly visible. Unnecessary information should be avoided where possible.

2. Keep software updated

CMS platforms, plugins, and other software should be kept updated regularly. Organizations should also check for security advisories related to the software they are using.

3. Review HTTP headers

HTTP response headers should be reviewed to check whether they are revealing unnecessary technical information about the server or application.

4. Review DNS records regularly

DNS records should be checked from time to time to make sure that only the required information and services are publicly exposed.

5. Properly configure and monitor the WAF

The WAF (ModSecurity) should be kept enabled and properly configured. It should also be monitored and tuned when required, since it can block basic or naive attacks.

6. Perform regular internal network discovery

Organizations should regularly scan their own networks to identify active devices and get a better understanding of what is connected to the network.

7. Investigate unknown or duplicate devices

Any unknown or unexpected device found during network discovery should be investigated. Devices showing the same MAC address should also be checked to make sure there is no configuration issue or unauthorized device.

8. Maintain network documentation

Network topology and device information should be properly documented and updated whenever changes are made to the network.

9. Perform security testing with authorization

Reconnaissance, scanning, and other security testing activities should only be performed on systems and networks where proper authorization has been given.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking learning program, I completed practical activities related to footprinting, reconnaissance, and network scanning.

In the footprinting activity, I used six different Kali Linux tools to collect information about the target domain. Through this activity, I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can be used to check HTTP headers, Wafw00f can help identify a WAF, and DNSRecon can provide additional information about DNS records.

In the network scanning activity, I used Zenmap to check my local network configuration and discover active devices on the network. I also collected IP and MAC address information and created a network topology. During the scan, I noticed an unusual duplicate-MAC finding, which showed me why unexpected results should be investigated instead of being ignored.

These activities helped me understand that information gathering is an important part of cybersecurity. Before performing any further security testing, a security professional can learn a lot about a system or network by carefully looking at publicly available information and the responses received from the network.

I also learned the importance of properly documenting security findings. A good cybersecurity report should clearly explain what was done, what was found, what the finding means, what risk it could create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and network scanning should always be performed within an authorized scope. All of these activities were carried out as part of my assigned educational cybersecurity lab.

8. Evidences Collected

Screenshots and raw output for each activity are collected here.

4.1 Footprinting evidence:

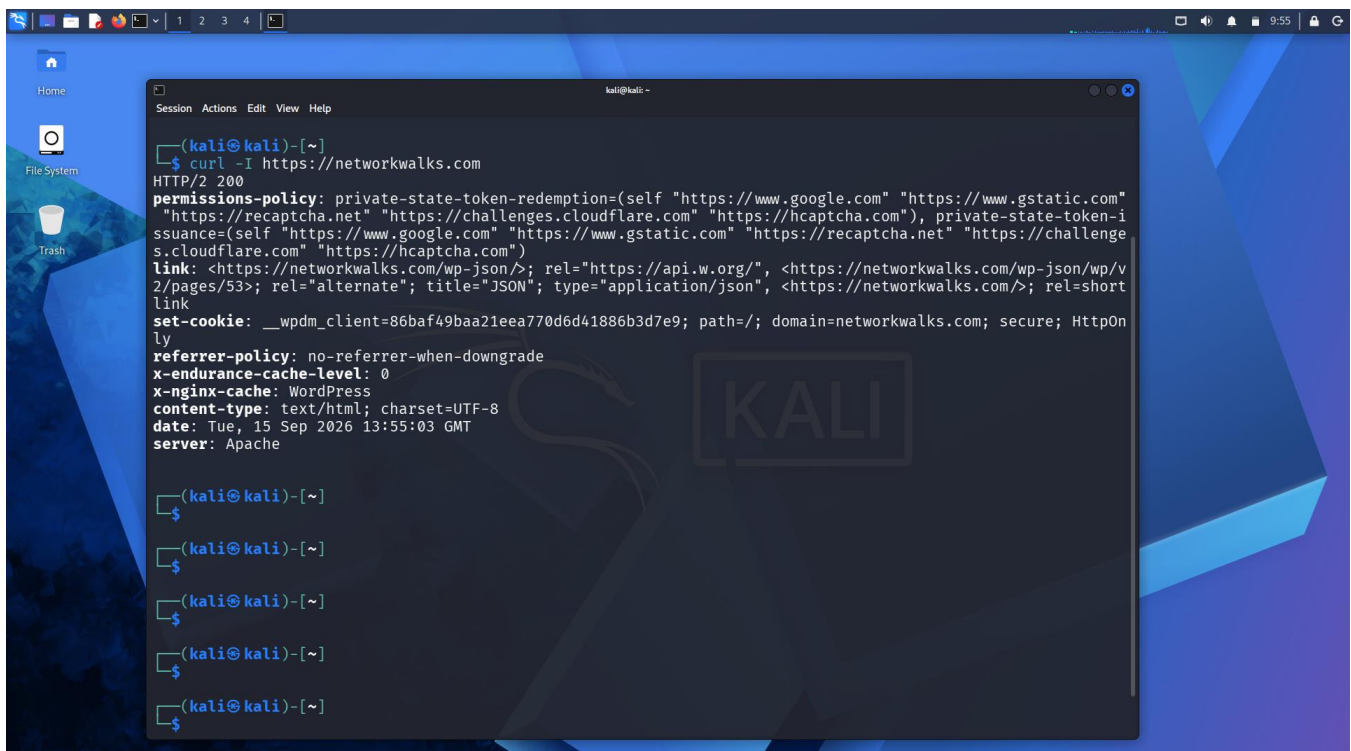
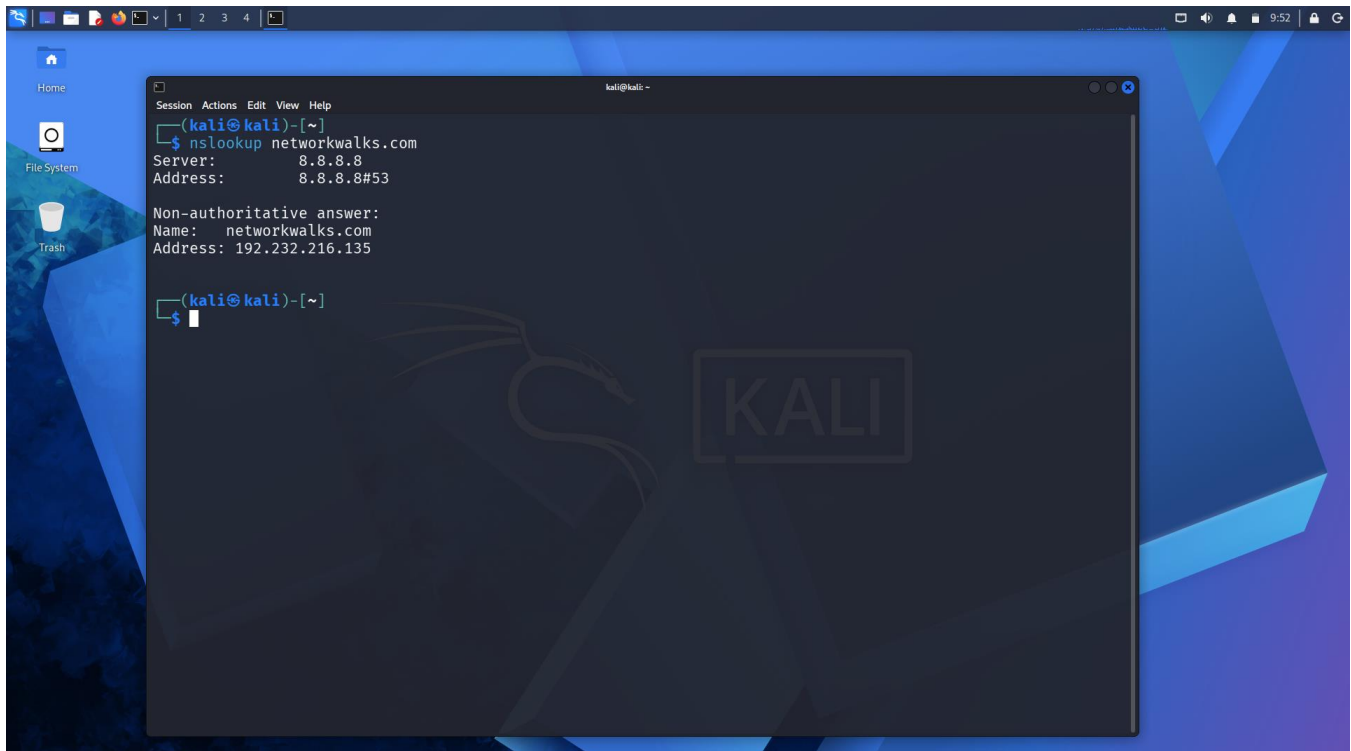
```
kali@kali:~$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-09-15T13:10:24Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
```

```
kali@kali:~$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&prtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&prtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]
```



```
Session Actions Edit View Help
kali@kali ~
(kali@kali)~$
(kali@kali)~$
(kali@kali)~$
(kali@kali)~$
(kali@kali)~$
(kali@kali)~$
(kali@kali)~$ wafw00f networkwalks.com

      ?
      ??
      ???
      ( _ ) _ ;
      \ _ / \ /

~ WAFW00F : v2.4.2 ~
~ Sniffing Web Application Firewalls since 2009 ~

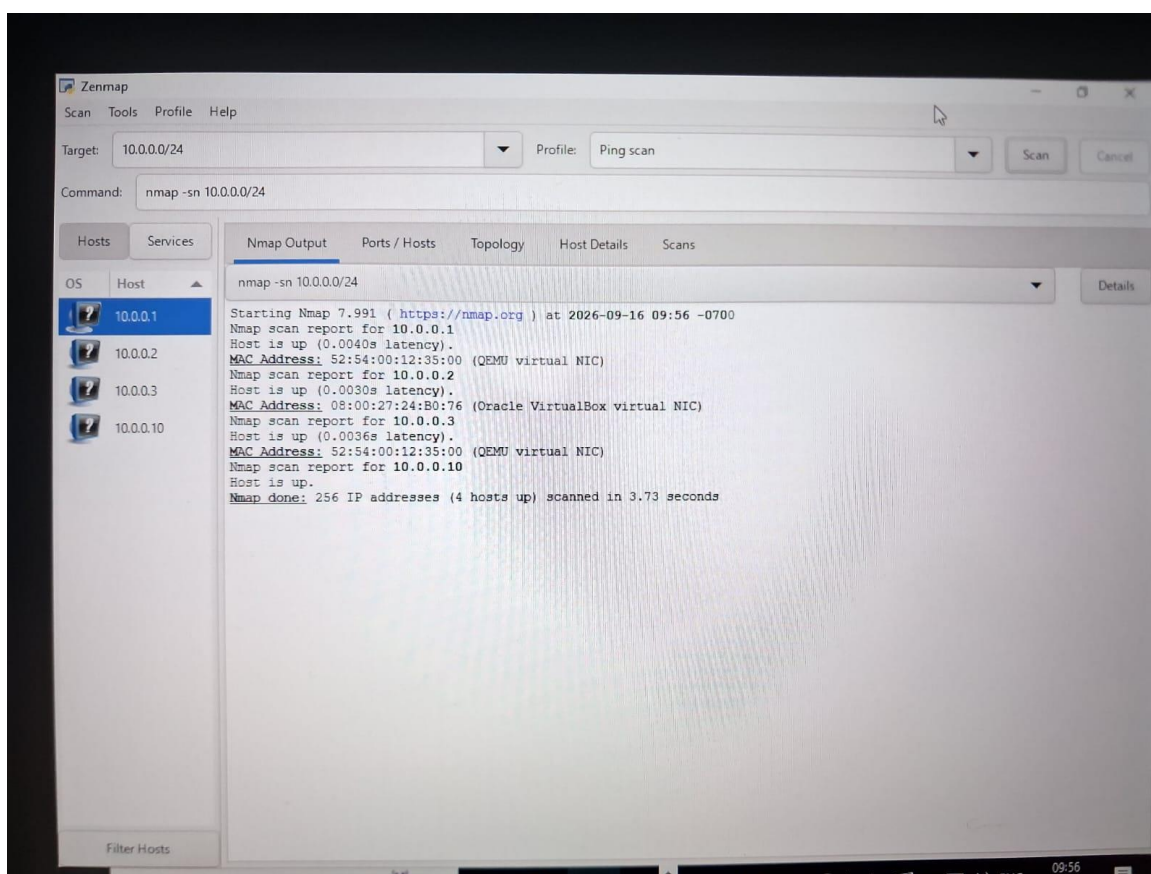
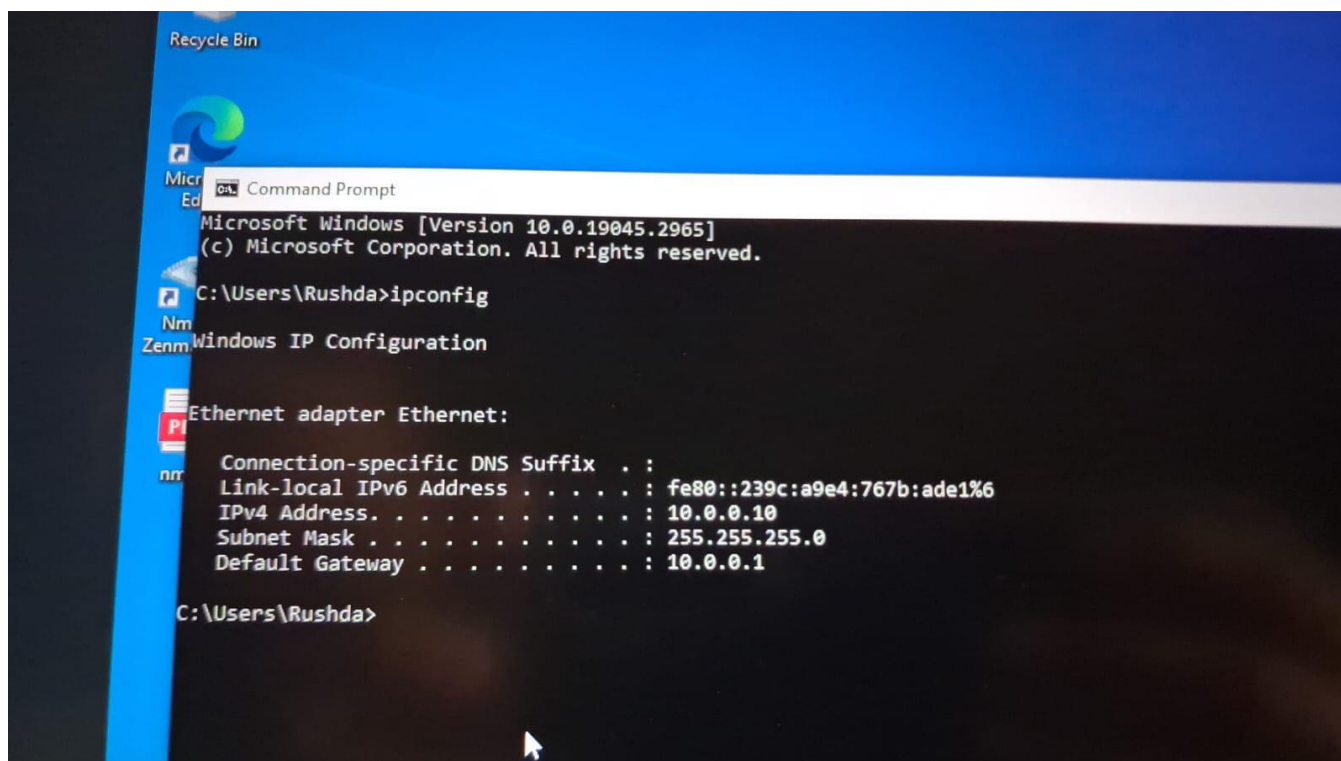
[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2

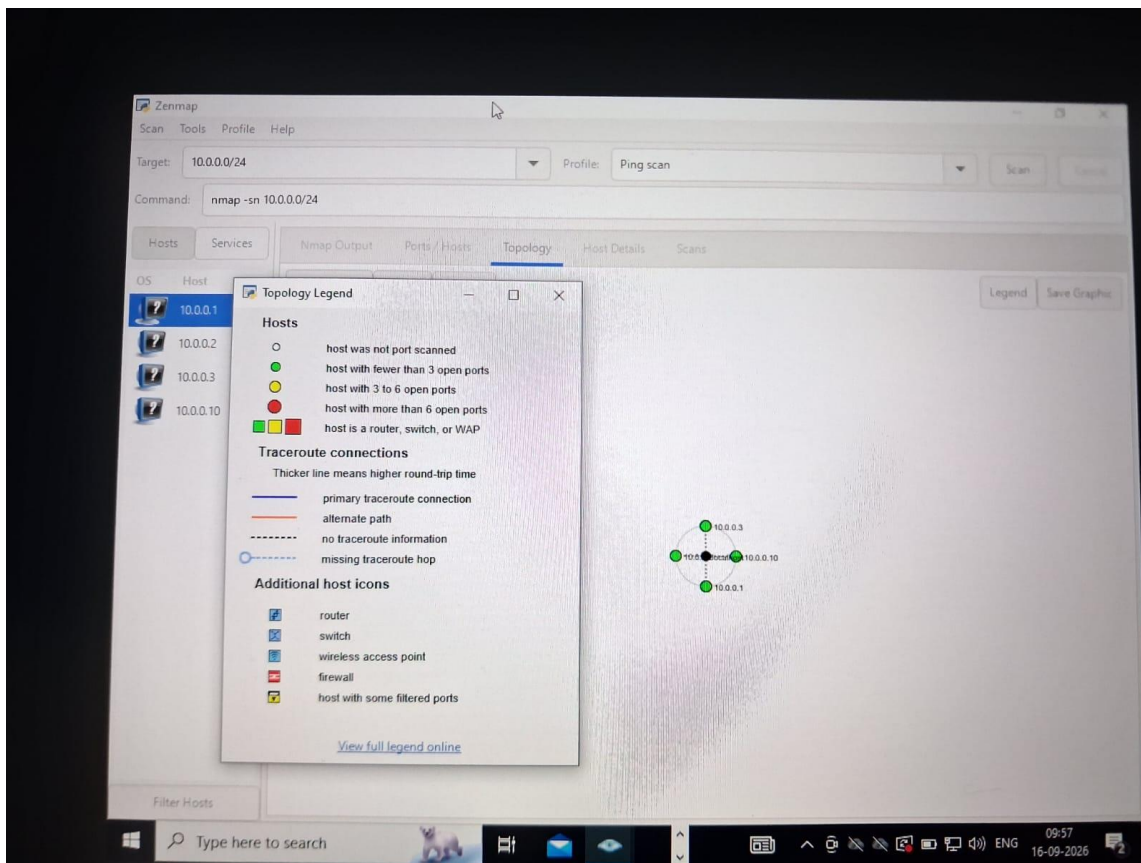
(kali@kali)~$
```

```
Session Actions Edit View Help
kali@kali ~
$ dnsrecon -d networkwalks.com
2026-09-15T09:58:50.883631-0400 INFO Starting enumeration for domain: networkwalks.com
2026-09-15T09:58:50.884214-0400 INFO std: Performing General Enumeration against: networkwalks.com...
2026-09-15T09:58:51.833668-0400 ERROR No answer for DNSSEC query for networkwalks.com
2026-09-15T09:58:52.180551-0400 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-09-15T09:58:52.513813-0400 INFO NS ns6135.hostgator.com 50.87.144.87
2026-09-15T09:58:53.172512-0400 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-09-15T09:58:53.172851-0400 INFO NS ns6136.hostgator.com 192.232.216.131
2026-09-15T09:58:53.779568-0400 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-09-15T09:58:54.683252-0400 INFO MX mail.networkwalks.com 192.232.216.135
2026-09-15T09:58:55.009094-0400 INFO A networkwalks.com 192.232.216.135
2026-09-15T09:58:55.870318-0400 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:webs
itewelcome.com ~all
2026-09-15T09:58:55.870742-0400 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizD
NVK4q75X-Ij-mjgEeg-UsYI
2026-09-15T09:58:56.443484-0400 INFO Enumerating SRV Records
2026-09-15T09:59:00.145452-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpane
l.net 184.94.203.14 443
2026-09-15T09:59:00.145913-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpane
l.net 184.94.204.11 443
2026-09-15T09:59:00.146086-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpane
l.net 184.94.203.9 443
2026-09-15T09:59:00.146288-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpane
l.net 184.94.204.14 443
2026-09-15T09:59:00.146416-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpane
l.net 184.94.204.9 443
2026-09-15T09:59:00.146532-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpane
l.net 184.94.203.11 443
2026-09-15T09:59:00.146644-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpane
l.net 184.94.204.15 443
2026-09-15T09:59:00.146930-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpane
l.net 184.94.203.15 443
2026-09-15T09:59:00.148652-0400 INFO 8 Records Found
2026-09-15T09:59:00.149599-0400 INFO Completed enumeration for domain: networkwalks.com

(kali@kali)~$
```

4.2 Network scanning evidence:





-End-

□ Author

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□ Project Information

Program Name: Cybersecurity program at Networkwalks | Week: 02